MZ UCA Cloud Academy

AWS Solutions Architect Associate (SAA)

Architecture Exercise 📐

Identity & Access Management (IAM)

Thinking Like a Cloud Governance Architect

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◈ EXERCISE OVERVIEW

In this exercise, you will move beyond memorizing AWS services and begin thinking like a Solutions Architect.

Your task is to design a secure Identity and Access Management (IAM) structure for a growing company using AWS.

◈ BUSINESS SCENARIO

Company Profile

Ubuntu Retail Group (URG) is expanding its operations across Southern Africa and has decided to migrate its infrastructure to AWS.

The company has four departments:

◆ Executive Management

◆ Finance

◆ Developers

◆ Security & Operations

The company currently employs 25 staff members and expects rapid growth over the next 12 months.

Management requires:

✔ Secure access control

✔ Least Privilege implementation

✔ High accountability

✔ Reduced security risk

✔ Scalable permission management

✔ Compliance-ready architecture

◈ PART 1 – IDENTIFY THE IAM USERS

Task

Create IAM Users for the following employees:

Employee	Department
Grace	Finance
Ndabezitha	Developer
Vusi	Developer
Boitumelo	Operations
Sarah	Executive Management
David	Security

Architecture Questions

1. Why should each employee have their own IAM User?

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1. What risks exist if all employees share one AWS account?

◈ PART 2 – DESIGN IAM GROUPS

Task

Design IAM Groups for the organization.

Complete the table below.

IAM Group	Purpose
____	____
____	____
____	____
____	____

Architecture Questions

1. Why are Groups better than assigning permissions directly to users?

1. Which department should receive the most permissions?

Why?

◈ PART 3 – ASSIGN PERMISSIONS

Task

Determine the minimum permissions required for each department.

Developers

Choose appropriate permissions:

☐ Launch EC2

☐ Stop EC2

☐ Read CloudWatch Logs

☐ Delete IAM Users

☐ View S3 Objects

☐ Modify Billing

Finance Team

Choose appropriate permissions:

☐ View Billing

☐ Create Budgets

☐ View Cost Explorer

☐ Delete EC2 Instances

☐ Manage IAM Policies

☐ View Cost Reports

Security Team

Choose appropriate permissions:

☐ Review IAM Access

☐ View CloudTrail Logs

☐ Enable MFA

☐ Audit Permissions

☐ Manage Security Controls

☐ Read Security Findings

Architecture Question

Which permissions would violate the Principle of Least Privilege?

◈ PART 4 – IAM POLICY DESIGN

Scenario

The Developers Group requires access to:

✔ Launch EC2 Instances

✔ Stop EC2 Instances

✔ View CloudWatch Logs

They should NOT be allowed to:

✖ Delete IAM Users

✖ Access Billing Information

Task

Answer the following:

1. Would this policy use Allow or Deny statements?

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1. Which AWS resources are being protected?

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1. Why should billing permissions remain restricted?

◈ PART 5 – ROLE-BASED ACCESS

Scenario

An EC2 web server must access an S3 bucket containing application files.

Task

Choose the best architecture solution.

☐ Store AWS Access Keys on the EC2 Instance

☐ Create a Shared IAM User

☐ Attach an IAM Role to the EC2 Instance

☐ Disable Security Controls

Architecture Question

Explain why your choice is the most secure option.

◈ PART 6 – LEAST PRIVILEGE REVIEW

For each scenario, determine whether Least Privilege is being followed.

Scenario A

Developer receives AdministratorAccess.

☐ YES

☐ NO

Explain:

Scenario B

Developer can start and stop EC2 instances only.

☐ YES

☐ NO

Explain:

Scenario C

Finance team can view billing but cannot modify security settings.

☐ YES

☐ NO

Explain:

Scenario D

Security team has access to CloudTrail and IAM auditing tools.

☐ YES

☐ NO

Explain:

◈ PART 7 – ENTERPRISE ARCHITECTURE DESIGN

Task

Draw your own IAM architecture.

Example:

Root Account

│

├──── Finance Group

│

├──── Developers Group

│

├──── Security Group

│

├──── Operations Group

│

└──── Executive Group

Roles

├──── EC2 Role

├──── Lambda Role

├──── Backup Role

└──── Cross-Account Role

Student Design

(Use this space to sketch your architecture.)

◈ ARCHITECT CHALLENGE

A new intern joins the Development Team.

What is the MOST GOVERNANCE-ALIGNED approach?

☐ Create a new IAM User and add them to the Developers Group

☐ Share another developer's credentials

☐ Give AdministratorAccess immediately

☐ Use the Root Account

Explain Your Answer

◈ EXECUTIVE REFLECTION

A Cloud Governance Architect must balance:

✔ Security

✔ Productivity

✔ Scalability

✔ Compliance

✔ Operational Efficiency

Reflection Questions

1. What is the biggest security risk in a poorly designed IAM environment?

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1. Why are IAM Roles preferred over Access Keys?

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1. Why is Least Privilege considered a foundational cloud security principle?

◈ ARCHITECT'S TAKEAWAY

A Solutions Architect does not ask:

"Can users access the system?"

A Solutions Architect asks:

"Should users access the system, and if so, what is the minimum level of access required?"

That mindset is the difference between administration and architecture.

MZ UCA PRINCIPLE

"Identity drives security.

Governance drives trust.

Least Privilege drives resilience."

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